Controlled Resolution Scaling Analysis for Transformer-Based Monocular Depth Estimation Under Edge Deployment Constraints

Jiyu Li
Department of Software Design
Technological University of the Shannon
Athlone, Ireland
A00316304@student.tus.ie

Cian Farrell
Department of Software Design
Technological University of the Shannon
Athlone, Ireland
A00293701@student.tus.ie

Priyanka Burkul
Department of Software Design
Technological University of the Shannon
Athlone, Ireland
A00335939@student.tus.ie

Taha Aflouk
Department of Software Design
Technological University of the Shannon
Athlone, Ireland
A00300601@student.tus.ie

Abstract — Monocular depth estimation offers a low-cost alternative to hardware-intensive sensing for assistive navigation, yet transformer-based architectures impose computational overhead that limits real-time edge deployment in safety-critical settings. This study evaluates whether deployment feasibility can be improved without architectural modification by isolating input resolution scaling as the sole optimisation variable under a locked protocol. A transformer depth model was assessed against a fixed 1920×1080 reference baseline, measuring ROI depth consistency and synchronised inference throughput. Although moderate downsampling preserves relative consistency, throughput remains effectively constant across resolutions, producing no Pareto improvement in the speed–accuracy trade-off. These results indicate that input scaling alone cannot bridge the real-time deployment gap, and that architectural efficiency is required.

Index Terms — Monocular Depth Estimation, Vision Transformers, Resolution Scaling, Edge Deployment, Assistive Navigation, Real-Time Inference.

I. Introduction

Monocular depth estimation infers 3D scene structure from a single RGB image and offers an affordable alternative to hardware-intensive navigation systems such as LiDAR or stereo cameras for visually impaired users [1], [2]. Although deep learning has improved global depth consistency, modern models impose significant computational constraints that limit real-time edge deployment.

Inference latency is safety-critical in assistive navigation. Human auditory response and gait termination require reaction time, and system latency exceeding approximately 200 ms reduces the available safety buffer [3], [4].

Vision Transformer (ViT) models such as DPT and MiDaS achieve superior depth accuracy compared to CNNs, but introduce substantial computational overhead [5], [6]. While existing optimisation strategies largely focus on model compression, controlled evaluation of input resolution scaling as an isolated deployment variable remains limited.

This study treats input resolution as the sole independent variable and evaluates its effect on inference throughput and Region of Interest (ROI) depth consistency under a locked protocol. Initial results show that moderate downsampling preserves consistency but does not improve throughput, indicating that resolution scaling alone cannot bridge the real-time deployment gap.

II. Literature Review

A. From Convolutional Networks to Transformers

Early monocular depth estimation systems predominantly employed Convolutional Neural Networks (CNNs) [5], [7]. Based on local receptive fields and hierarchical feature extraction, these architectures are computationally efficient and suitable for higher frame-rate inference on constrained hardware. However, CNN-based approaches can struggle to capture long-range spatial dependencies, leading to weaker global scene consistency.

Vision Transformer (ViT) architectures substantially shifted this paradigm. Models such as the Dense Prediction Transformer (DPT) use self-attention to model global context across the image [6], achieving lower Root Mean Square Error (RMSE) than CNN baselines in comparative studies [5], [6]. This accuracy improvement, however, introduces increased computational overhead due to

attention mechanisms and architectural complexity [6], [7], often resulting in sub-real-time performance on edge-constrained devices [6], [7].

B. Efficiency Strategies for Edge Deployment

To address deployment constraints, recent work has focused on architectural optimisation and model compression [7]. Knowledge distillation transfers representations from large transformer backbones to lighter student models, while quantisation and pruning reduce parameter precision and computational load [1], [7].

Despite these efforts, relatively few studies isolate input resolution scaling as a controlled optimisation variable in depth estimation pipelines [5], [7]. In CNN-based systems, runtime typically decreases proportionally with spatial downsampling [5]. Transformer-based models, however, may exhibit weaker scaling due to fixed computational costs associated with preprocessing, patch embedding, and attention mechanisms [6]. This raises a key deployment question: can input resolution scaling alone meaningfully improve real-time feasibility under edge constraints?

C. Latency and Safety in Assistive Navigation

In assistive navigation, latency must be treated as a safety constraint rather than a flexible performance metric. Human auditory response time is approximately 0.24 seconds, and gait termination requires additional reaction time [3], [4]. When system latency exceeds ~200 ms, the available safety margin decreases, increasing collision risk [1], [2].

Practical systems should therefore target inference speeds above approximately 15 FPS to maintain responsive guidance under safety constraints [1]–[4]. Although the literature frequently reports improvements in depth accuracy, fewer studies evaluate whether such models satisfy strict real-time deployment thresholds on edge hardware, particularly within ground-level safety-critical regions [1], [2].

III. Test Configuration

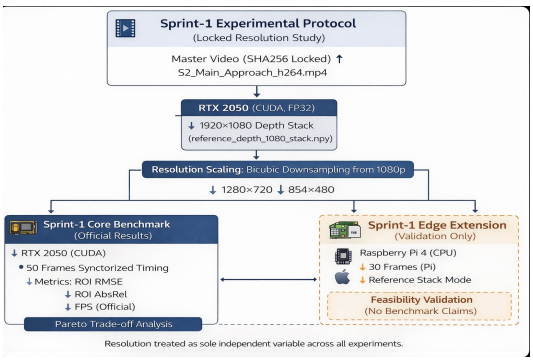

Figure 1. Locked experimental protocol showing input resolution as the sole independent variable. The RTX 2050 branch provides the official benchmark, while Raspberry Pi 4 and Apple M1 provide feasibility and cross-backend validation.

All experiments followed a strictly locked, reproducible protocol in which input resolution was the only independent variable. The master video (S2_Main_Approach_h264.mp4) was SHA256-locked to prevent dataset drift. A 1920×1080 reference depth stack (reference_depth_1080_stack.npy) was generated once on an RTX 2050 (CUDA, FP32) and reused in file-reference mode for all non-reference runs to ensure consistent metric comparison across resolutions and devices.

The Region of Interest (ROI) was defined as the bottom 40% and central 60% of each frame to reflect safety-critical ground-level regions. Accuracy was evaluated using ROI RMSE and ROI AbsRel ($\varepsilon = 1e{-}6$). Bicubic interpolation was used for all resizing operations. CSV outputs were treated as the single source of truth and were not manually modified.

A. Core Benchmark (RTX 2050)

The official benchmark was executed on an NVIDIA RTX 2050 (CUDA backend only) at three input resolutions: 1920×1080 (reference baseline), 1280×720, and 854×480. Performance was measured over 50 frames using forward-pass-only execution with CUDA-synchronised timing. Only RTX measurements are used for throughput claims and Pareto trade-off analysis.

B. Edge Feasibility Extension (Raspberry Pi 4)

A feasibility evaluation was conducted on a Raspberry Pi 4 under CPU-only execution at 1280×720, 854×480, and 640×360. The 1920×1080 configuration was not executed due to infeasibility. These results are reported strictly as feasibility context and do not redefine the official benchmark.

C. Cross-Backend Verification (Apple M1)

Metric consistency was verified on Apple M1 (MPS) at 1920×1080, 1280×720, and 854×480 using the same locked RTX 1080p reference stack. M1 results are for validation only and are not used for performance benchmarking.

IV. Results

A. Core benchmark (RTX 2050 — Official Benchmark)

This phase evaluates controlled resolution scaling under CUDA execution with input resolution as the sole independent variable. **Figure 2** shows ROI RMSE behaviour versus megapixel count. Reducing resolution from 1920×1080 to 854×480 produces progressive degradation in reference-aligned depth consistency. The increase is monotonic but non-linear, indicating that degradation does not scale proportionally with pixel count. The 1920×1080 condition has RMSE = 0 by construction because it is the fixed reference baseline.

AbsRel trends mirror RMSE behaviour, supporting metric consistency and stability under the locked evaluation pipeline.

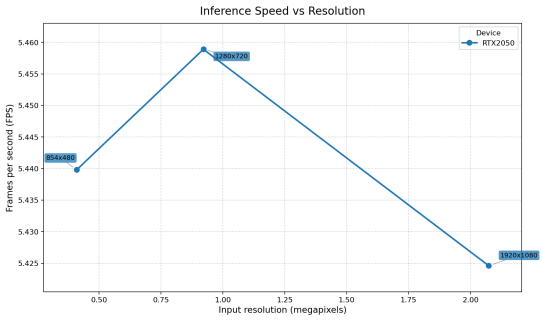

Figure 3 reports RTX inference speed under CUDA-synchronised timing across 50 measured frames. Despite an approximately 80.2% pixel reduction (1080p → 480p), throughput remains effectively constant (~5.42–5.46 FPS) with low variance, indicating runtime is dominated by architectural overhead rather than pixel count.

This aligns with transformer dense-prediction complexity characteristics reported in the literature [6].

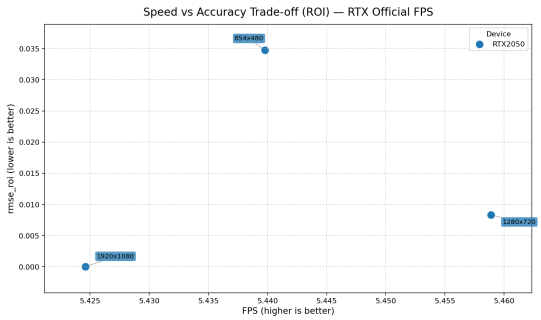

Figure 4 visualises the speed–accuracy trade-off using official RTX FPS values. Because FPS remains constant while consistency degrades with downsampling, no configuration yields a Pareto improvement within the evaluated resolution band. This demonstrates that input scaling alone does not shift the speed–consistency frontier for the tested architecture

B. Core Edge Extension (Validation & Feasibility Layer)

This extension evaluates feasibility and cross-device behaviour on non-CUDA backends (Raspberry Pi CPU and Apple M1 MPS) and does not replace the core timing study.

Raspberry Pi 4 (CPU-only). Throughput remains < 0.3 FPS even at 640×360. Moderate downsampling (720p, 480p) preserves ROI RMSE at approximately ~0.20, while 640×360 introduces abrupt degradation (RMSE ~0.730), indicating loss of spatial coherence within the safety ROI. Real-time deployment is infeasible.

Apple M1 (MPS). Throughput improves relative to Pi but remains well below the 15 FPS threshold. ROI consistency trends mirror Pi behaviour, indicating reference alignment and no device-specific instability. M1 verification was performed at 1080p, 720p, and 480p using the locked RTX 1080p reference stack; 640×360 was evaluated only on Raspberry Pi in this phase.

C. Discussion

1) Cross-Device Consistency

M1 verification at matched resolutions produced ROI trends consistent with the RTX core benchmark, supporting reference alignment under a non-CUDA backend. Pi results are reported strictly as feasibility context and show substantially lower throughput and higher error, particularly at 640×360.

2) Resolution–Accuracy Threshold Behaviour

Between 1080p and 480p, ROI consistency remains stable. At 640×360, degradation becomes abrupt rather than gradual, suggesting a spatial granularity threshold below which transformer attention mechanisms lose structural coherence within safety-critical regions.

3) Empirical Validation of Literature Claims

Existing literature highlights a trade-off between transformer-based depth accuracy and deployment feasibility. This phase empirically quantifies that boundary under controlled conditions. While transformer models maintain strong structural consistency, they fail to approach real-time performance under the locked configuration.

4) Safety-Critical Implications

Inference latency exceeding approximately 200 ms reduces the available reaction window during walking [3], [4]. Although ROI consistency can be preserved under moderate downsampling, throughput remains well below safety-relevant real-time thresholds. Efficiency must therefore be prioritised in deployable assistive navigation systems

5) Resolution–Performance Trade-off

Across the evaluated resolution band, ROI error increases while RTX throughput remains effectively constant despite substantial pixel reduction. No Pareto improvement is observed, confirming that input scaling does not shift the speed–consistency frontier under the evaluated transformer architecture.

V. Conclusion/Future Work

This study demonstrates that controlled input resolution scaling does not improve real-time feasibility for transformer-based monocular depth estimation under a locked reference protocol. While spatial consistency degrades progressively with downsampling, throughput remains constrained by architectural overhead rather than pixel count.

Edge validation further confirms that CPU-only deployment is infeasible and that aggressive downsampling introduces instability within safety-critical regions.

Future work will preserve the locked dataset, ROI definition, and timing protocol while introducing model architecture as the sole independent variable to determine whether architectural optimisation can meaningfully shift the speed–consistency trade-off.

VI. References

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